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RESEARCH ARTICLE



Artificial Intelligence Research in Social Media Marketing: A Review, Synthesis and Research Avenues

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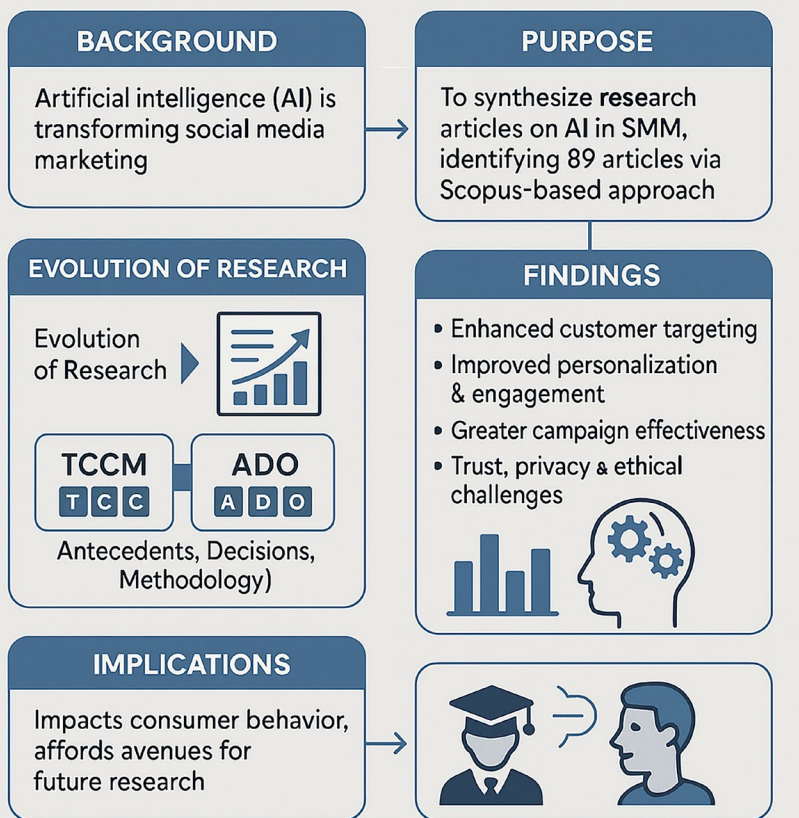
ABSTRACT

Artificial Intelligence (AI) is changing how businesses use social media for marketing by making it more personalized, efficient, and data-driven. Its expanding role in Social Media Marketing (SMM) emphasizes the critical need for this systematic literature review (SLR) to uncover its evolving impact and potential. This study synthesizes findings from research articles on AI usage in social media marketing, where 89 articles were identified systematically and extracted from Scopus-based literature. This review highlighted the evolution of research in the domain using the SPAR-4-SLR methodology. Guided by the TCCM (Theory, Context, Characteristics, Methodology) and Antecedents-Decisions-Outcomes (ADO) frameworks, the study explores what drives AI use in SMM, how decisions are shaped, and the resulting various outcomes. The findings show that AI strengthens various aspects of SMM, particularly customer targeting, personalization, engagement, and campaign effectiveness, while introducing challenges related to trust, privacy concerns, and ethical use of AI technologies. This review provides useful insights for academics and practitioners by describing how artificial intelligence used in SMM impacts consumer behavior. It emphasizes the need for further cross-cultural, industry-specific, and advanced methodological research to support responsible and effective AI in SMM initiatives.

KEYWORDS

Artificial intelligence; social media marketing; systematic literature review; TCCM framework; ADO framework

GRAPHICAL ABSTRACT



Introduction

The use of cutting-edge technologies in the digital era has initiated a transformative wave of digitalization across industries (Labib, 2024; Smailhodzic et al., 2025). Industry 4.0 is a revolution that allows businesses to integrate advanced digital technologies (Črešnar & Nedelko, 2020). According to Ali and Johl (2023), the term “4.0” is comprehensive and may be used to group a range of technologies, including the Internet of Things (IoT), Artificial Intelligence (AI) and big data. AI is a technologically advanced system that evaluates service conditions in real-time by leveraging both digital and physical data, offering personalized recommendations, alternatives, and explanations to customer inquiries or issues (Xu et al., 2020). In the rapidly shifting landscape of consumer behavior and market dynamics, employing AI in marketing research is vital (Ali, Balta et al., 2023). In general, AI is a revolutionary technology that has the power to replace established standards and transform the production and consumption of goods and services (Kim et al., 2025). Industry 6.0 is a completely automated system that uses simple clear instructions to regulate product design, production, and assembly through coordinated robotic swarms and generative artificial intelligence.

Further, AI tools are highly efficient in tracking, analyzing social media activity (Sterne, 2017) and understanding what people are talking on these platforms (Capatina et al., 2020). According to marketing experts, AI has significant advantages for consumers as social media usage increases, it rises the need for AI that comprehends consumer wants (Pitardi & Marriott, 2021). In particular, the social media AI market is predicted to grow to US\$12.43 billion by 2030 (Maximize Market Research Pvt Ltd, 2024).

Social media marketing research has advanced to new prospects to reflect smart consumers and more intellectual businesses facilitated by AI. Prior studies have used interpretative analysis and systematic literature review to synthesize marketing research and AI (Chintalapati & Pandey, 2022; Labib, 2024; Mariani et al., 2022). The majority of past studies have examined marketing as a whole, but social media marketing (SMM) is a specialized

area that requires study on its own. SMM is distinctive for its user-generated content, real-time engagement, and sophisticated algorithms making it more complicated but also more engaging than traditional online marketing (Chyrak et al., 2024). At the same time, SMM is also associated with a number of issues, including data privacy, honesty of influencers, excessive dependence on algorithms, and the dissemination of false information (Żyminkowska & Zachurzok-Srebrny, 2025).

Although AI has been commonly studied in marketing, its use in SMM is different and deserves special focus. AI is mostly used in traditional online marketing contexts for structured data and firm-centric tasks like targeting, forecasting, and consumer segmentation (Verma et al., 2021). Reviews that have already been written frequently discuss AI in marketing in general, which leads to fragmented insights and contradictory conclusions (Marvi et al., 2025). Thus, they may not address whether AI-driven personalization increases engagement or increases distrust and privacy concerns. The real-time, interactive, and consumer-driven nature of SMM, on the other hand, necessitates that AI analyze tremendous amounts of unstructured multidimensional information while interacting directly with online communities (Lyulyov et al., 2025). As a result, there are opportunities like credibility, advertisement, trust, image richness and issues with ethical use, privacy, and authenticity that are less noticeable in larger marketing settings (Żyminkowska & Zachurzok-Srebrny, 2025). Despite all of these noteworthy mentions, none of the evaluations fully address the corpus of review research on AI in SMM. However, significant comprehensive progress has been made in understanding AI's role and application in marketing. But there is no such comprehensive analysis that focuses primarily on how AI is used in SMM. So, in light of the identified research gap, the following research questions have been addressed by this review:

RQ1: *What do we now know about the dynamics of AI usage in SMM, including its antecedents, decisions, and outcomes?*

RQ2: *What theories, contexts, characteristics, and methods have been examined in existing AI in SMM studies?*

RQ3: *What major implications could be drawn from the current research on AI in SMM?*

RQ4: *What future directions should researchers consider in the field of AI in social media marketing?*

Recent years have seen a significant increase in literature on AI and social media, with 86% of publications published in the past five years (January 2020–May 2025). This increase suggests that the discipline is ready for a hybrid systematic review that can cover the entire topic. Our evaluation takes Lim et al. (2022) approach, combining the SPAR-4-SLR technique (Paul et al., 2021) with the structured TCCM-ADO framework (Paul & Rosado-Serrano, 2019). This approach combines AI with SMM research to thoroughly understand the conceptual and theoretical foundations, including theory, context, characteristics, and methodology (TCCM), providing practical implication for marketers and recommending future research areas from the TCCM perspective.

Overview of artificial intelligence in social media marketing

AI is defined as “the use of computer algorithms to imitate the capacities of people, including thinking, executing physical operations, and feeling” (Huang & Rust, 2021). Earlier studies highlight how useful AI is for enhancing SMM strategies (Basri, 2020). Emerging AI technologies used by SMM analyze consumer-generated content, offering valuable insights into brand perceptions and attributes (Brüns & Meißner, 2024). AI has the potential to improve effective managerial practices by enabling a variety of business activities, including marketing, and improving company performance (Basri, 2020). The integration has shifted the marketing paradigm from conventional to advanced digital strategies (Henry, 2019).

Despite the notable rise in AI research, a considerable gap persists in thoroughly examining the present and prospective applications of AI approaches in SMM strategies (Labib, 2024). Some studies claim that the automation of SMM leads to the increased use of AI, with the following trend being further noted to help in tracking and comprehending all aspects of user behavior (Zheng et al., 2018). AI plays a substantial role in

SMM (Basri, 2020; Köhl et al., 2020) by providing cutting-edge technology that positively impacts various marketing efforts. Recent research, from both developed and developing nations, highlights the effectiveness of AI in SMM strategies (Basri, 2020). Given its considerable influence on marketing activities, SMM has emerged as one of the most important elements of contemporary marketing strategies (Shareef et al., 2019).

Methodology

A SLR provides an extensive and well-organized summary of the existing body of knowledge regarding a specific subject, idea, or area (Barta et al., 2025). The accuracy of the research process is increased, and the possibility of researcher bias in article selection is greatly reduced when using the SLR technique compared to the standard review (Vrontis et al., 2021). To synthesize the body of available research, this study uses a systematic review methodology. A systematic review follows a clear and replicable procedure that includes specified search strategies, inclusion and exclusion criteria, and a thorough evaluation of research, in contrast to traditional narrative reviews, which frequently offer descriptive overviews (Paul et al., 2021). By ensuring a thorough and objective synthesis, such rigorous methodology provides deeper insights into trends and gaps in the literature. For instance, Limbu and Huhmann (2022), who utilized it to investigate ethical difficulties in pharmaceutical marketing, and Sinha et al. (2025), who used a systematic review to decipher the relevance of avatars in marketing. In keeping with this history, the current review enhances its contribution to the field by highlighting understudied areas, integrating existing information, and suggesting future study possibilities.

To investigate issues such as “Exactly what we know, the way we know it, and in which direction we should be heading?” concerning AI and SMM, we opted for a framework-based review approach for the study conducted. The TCCM framework was adopted for the framework-based approach. In an attempt to make our findings easily understood, we will list the antecedents and outcomes of AI-SMM.

SPAR-4- SLR

To gather data, the SPAR-4-SLR methodology (Figure 1) is employed in the present research. According to Paul et al. (2021), it contains three primary stages: assembling, arranging, and assessing publications.

Assembling

The sub-stages of the assembling stage of the review protocol are identification and acquisition. The identification involves a definition of the research area, source quality and source type. The

topic of “AI in social media marketing” is highlighted in the comprehensive assessment of the literature. Additionally, the quality of the source was assured by taking the ABDC Journal Quality list (2022) and the Scopus Journal list (2023) into consideration. Furthermore, in the acquisition phase, the analysis was conducted without regard to a time frame, and on May 2025, a database search was carried out in electronic databases (Scopus). Scopus was used for data collecting because of its extensive coverage and rigorous indexing (Materla et al., 2019). While Web of Science (WoS) delivers clearer data, Scopus has

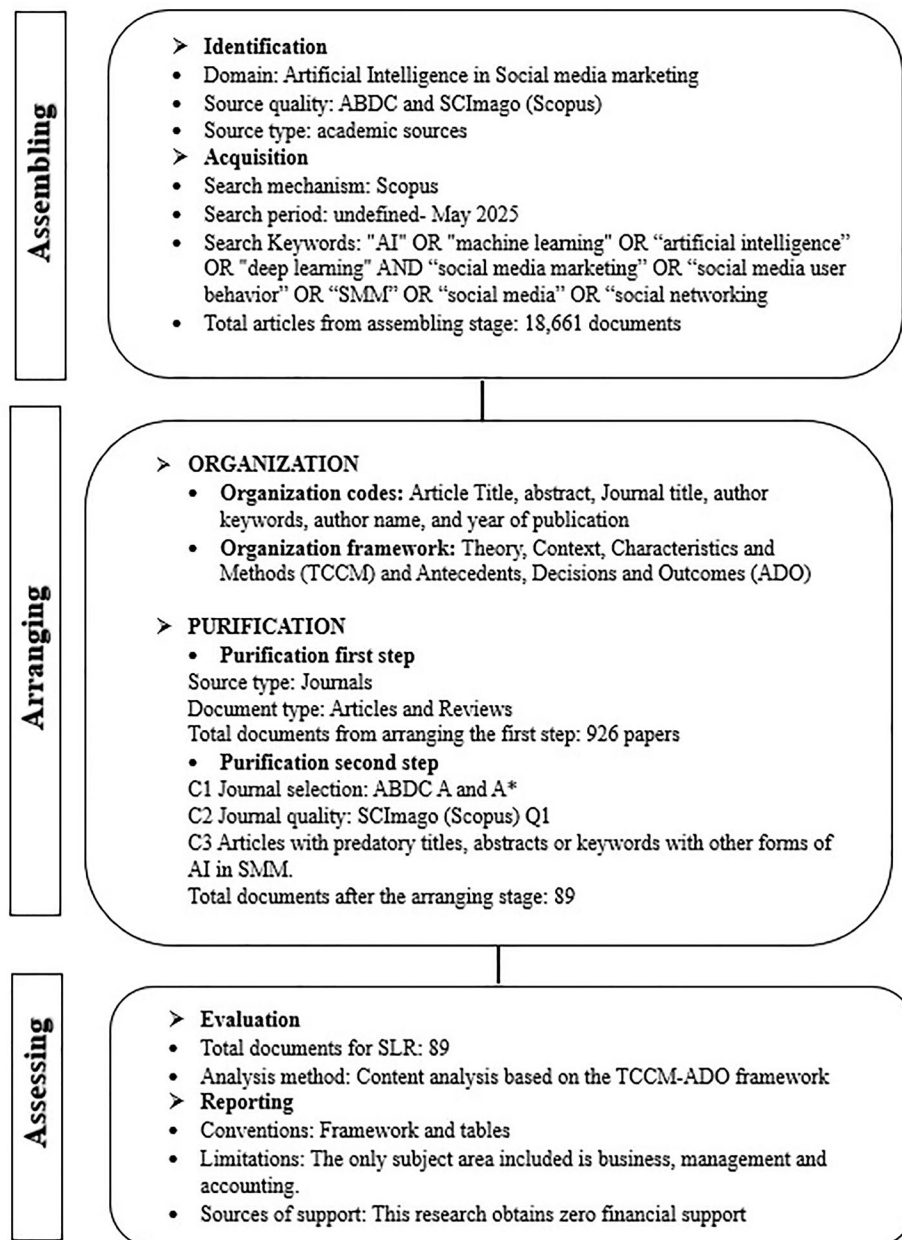


Figure 1. The SPAR-4-SLR protocol-based systematic literature review process.

around 60% greater coverage. Furthermore, to prevent data homogenization difficulties, SLRs in the social sciences frequently use a single database (Galati & Bigliardi, 2019). Based on the study's aim a specific research string was created using the keywords such as “deep learning” OR “machine learning” OR “AI” OR “artificial intelligence” AND “social media marketing” OR “social media user behavior” OR “SMM” OR “social media” OR “social networking” was refined using terms from earlier study and thesaurus and experts also review these keywords to ensure their relevance. A total of 18,661 studies were found in Scopus using these search keywords.

Arranging

The purification and organization substages were part of the arranging phase of this review. Article data were categorized for organization based on the author's name, country of affiliation, title, abstract, and keywords. To examine different aspects of the literature, the TCCM framework was chosen. Only articles from the “Business, Management, and Accounting” subject category in Scopus were chosen for the first purification phase. Only reviews and scholarly publications written in English were included, by Lim et al. (2022) and Fetscherin and Heinrich (2015). In this first stage, 926 papers that met the inclusion criteria were found.

The second stage of the selection process was methodical and focused on the goals and research questions of the study (Tranfield et al., 2003). Title, year, authors, source, abstract, keywords, and inclusion status were recorded in an Excel file for each manuscript, with C1–C3 denoting exclusion grounds. First, only journals ranked A and A* on the ABDC journal quality list were kept (C1). Then, following Chevtchouk et al. (2021), Quartile 1 journals were chosen from SCImago's 2023 index (C2). Finally, titles, abstracts, and keywords were used to filter articles (C3). There were 89 documents in the final dataset.

Assessing

The assessment stage of the SPAR-4-SLR procedure involves reporting the results and evaluating the literature that was selected. The peer-reviewed

papers that were part of our dataset were subject to a content analysis for assessment purposes. A framework-based technique, like TCCM-ADO, was employed for choosing the above articles (Wolf, 2023). After the articles about AI usage in SMM or similar subjects were selected, each one was thoroughly examined by providing in-depth knowledge that could advance our understanding of AI in SMM. Lastly, we offered a framework that should be helpful to both researchers and practitioners to give a comprehensive understanding of AI in SMM.

General information

Journal of publication

The publications were categorized to determine the publication sources and their impact on AI in SMM research. According to the ABDC 2024, 30 (33.70%) of the 89 publications were published in A* journals, with 59 (66.29%) appearing in A-rated journals. This is particularly interesting among prominent journals publishing AI in SMM research. Notable publications are Technological Forecasting and Social Change ($n=10$), European Journal of Marketing ($n=6$), Journal of Retailing and Consumer Services ($n=5$), Journal of Business Research ($n=5$), and International Journal of Information Management ($n=4$). Annexure A shows all the journals that have publications on this topic.

Theory, context, characteristics, methods (TCCM)

Several academics have gained important new insights from the notable studies on AI in SMM. However, a dearth of research exists that highlights all aspects of AI in SMM, which could motivate businesses and consumers to address the problem at a faster pace given the country and industrial settings.

Theory

Theoretical frameworks are essential for understanding the integration of AI in SMM. In our review of 89 research, we found out 14 theories used to explain the domain. These theories not only give conceptual clarity, but also aid in the interpretation of customer behavior, platform dynamics, and AI acceptance in social media

marketing environment. Various psychological theories, like social comparison theory (Festinger, 1954) and social exchange theory (Blau, 1964), are used to check the psychological states as well as emotional and motivational response of consumers from AI usage in social media platforms. For example, Ameen et al. (2022) and Fang and Hu (2018) employed Social Comparison Theory to explain how interacting with AI-curated information through social media affects self-perception and purchase intentions *via* peer benchmarking. Similarly, Cloarec et al. (2024) and Saad and Choura (2022) used Social Exchange Theory to show that perceived value from personalized AI interactions increases users' propensity to engage with branded material, emphasizing the importance of reciprocal value in AI-SMM engagement.

Also, Behavioral theories such as SOR Mehrabian and Russell (1974), the Uncanny valley theory (Mori, 1970), and Engel-Kollat-Blackwell (EKB) theory (Engel et al., 2001) are

applied in the studies focusing on how AI stimulus affects behavioral and emotional responses through SMM. They often overlook the influence of internal thoughts and emotions in favor of emphasizing how external stimuli and responses shape behavior of social media consumers. As indicated in Table 1, the Uncanny Valley theory (Mori, 1970) has been applied in three studies to anticipate how individuals categorize the unknown, such as robots or humans, using visual cues. According to these studies, including social cues and real-life aspects can help lessen perceptions of uncanniness. Furthermore, Cognitive theories such as attribution theory (Folkes, 1988) and cognitive dissonance theory (Festinger, 1957) are used to study how social media users interpret AI actions and justify their own choices. In this research, two studies used attribution theory (Folkes, 1988) to investigate how people rate the attractiveness of others based on their efforts. According to attribution theory, users of social

Table 1. Theories used in AI-SMM literature.

Theories	No. of articles	%	Implementation in AI-SMM research	Exemplary studies
Uncanny Valley Theory (Mori, 1970), (Mori et al., 2012)	3	15.7	Demonstrating Social Media customers' experience of incompetence in comparison to highly humanized products	Gutuleac et al. (2024), Kim and Park (2024), and Filieri et al. (2022)
Attribution Theory (Folkes, 1988)	2	10.5	Evaluate individual's attractiveness based on the perceived effort invested in their appearance.	Smith et al. (2022) and Fang and Hu (2018)
Social Exchange Theory (Blau, 1964)	2	10.5	Describe the psychological procedure that looks at how users are motivated to provide information for personalization is affected by their activity on social networking sites.	Cloarec et al. (2024) and Saad and Choura (2022)
Social Comparison Theory (Festinger, 1954)	2	10.5	Interacting within a social network, individuals can shape others' opinions or behaviors by simplifying their decision-making process, such as by narrowing down viable alternatives.	Ameen et al. (2022) and Fang and Hu (2018)
Social Support Theory (Cohen & Wills, 1985).	1	5.2	Examines several social support-related topics (such as consumer AI experience, customer satisfaction and social media engagement) as these light up interactions between humans and artificial intelligence.	Bilal et al. (2024)
Complexity Theory (Woodside, 2014)	1	5.2	Improves knowledge of the dynamics influencing social media user choices on utilizing virtual influencers.	Shao (2024)
Voluntary Disclosure Theory (Campbell et al., 2001)	1	5.2	Usage of AI tools to create and share before-and-after images, and infographics to communicate efforts as well as their results on social media platforms.	Bai et al. (2024)
Signal Transmission Theory (Connelly et al., 2011)	1	5.2	Exploring how environmental information disclosure can capture analysts' attention, which can influence investor market reactions	Bai et al. (2024)
Cognitive Dissonance Theory (Festinger (1957)	1	5.2	Describing how the conflict between convenience and privacy fuels their technology dissonance and reliance on social networking sites.	Wang et al. (2023)
Stimulus-Organism-Response (SOR) Mehrabian and Russell (1974).	1	5.2	Explaining individual decision-making behavior, the SOR model helps address the under-exploration of users' internal states on social media platforms.	Singh et al. (2023)
User and Gratifications Theory (Luo, 2002)	1	5.2	Understand how user's characteristics influence application usage, electronic word-of-mouth and product purchases.	Simay et al. (2023)
Engel-Kollat-Blackwell (EKB) Theory (Engel et al., 2001)	1	5.2	Expanding the model to include how artificial intelligence on Instagram influences consumers' purchase decisions for fashion retailers.	Yeo et al. (2022)
Grounded Theory (Glaser & Strauss, 1967)	1	5.2	Seeking to hire staff members as well as competent people whose behaviors and values align with their mission.	Allal-Chérif et al. (2021)
Shannon Information Entropy Theory (Shannon, 1948)	1	5.2	Provides an extensive collection of techniques for assessing information about unstructured observation occurrences.	Yang et al. (2021)

media want to comprehend the behaviors of others. In contrast, developmental theory, like grounded theory, is used to share a genuine commitment to the organization's purpose, fostering a cohesive and motivated workforce. This has been used in exploratory research to get insights from qualitative data, notably for recognizing changing user expectations and opposition to AI integration in SMM. Other theories used in AI and SMM research include complexity theory, the SOR model, cognitive dissonance theory, and user and gratification theory. These are also used in studies to explore users' motivation in AI environments.

Context

When defining the various circumstances that define AI in SMM, taking context into account is useful. This review explores the two contexts, industry and country, that form the basis of our

analysis of 89 studies. The developed economies around the world have been the focus of most research on social media and AI. Table 2 shows that most reviewed studies were conducted in North America (31 studies, or 36.04% of all studies) and Asia (21 studies, or 26.22% of all studies). Multiple studies examine social media and AI regardless of location, and increasingly, more studies have recently shifted their focus to European nations ($n=19$). According to the findings, most studies that have been published were done in a single country. Still, a few global perspective studies ($n=28$) cover multiple nations simultaneously. Research conducted in a single nation can only offer restricted understanding because its conclusions cannot be directly compared to those of other countries. The findings indicate that the maximum number of research that has been published was conducted in a single nation.

Table 2. Countries investigated in AI-SMM research.

Country	No. of studies	%	Exemplary studies
<i>America</i>			
North America (Canada, USA)	31	29.52	Brüns & Meißner (2024), Kozinets and Seraj-Aksit (2024), Ku et al. (2024), Oh and Ki (2024), Fu and Fisher (2023), Hu et al. (2023), Oravec (2023), Wang et al. (2023), Xie et al. (2023), Aljedaani et al. (2022), Gunarathne et al. (2022), Hartmann et al. (2021), Li et al. (2023), Meena and Kumar (2022), Philp et al. (2022), Sands et al. (2022), Van Esch and Stewart Black (2021), Van Esch et al. (2021), Hayes et al. (2021), Bunker (2020), Kietzmann et al. (2021), Obschonka et al. (2020), Wang et al. (2020), Buhalis and Sinarta (2019), Overgoor et al. (2019), Fang and Hu (2018), Cui et al. (2018), Meire et al. (2017), Moncrief (2017), Bell et al. (2014), Kwok & Yu (2013)
South America (Brazil)	1	1.00	Allal-Chérif et al. (2024)
<i>Europe</i>			
UK	11	10.47	Cloarec et al. (2024), Gutuleac et al. (2024), Shao (2024), Xu et al. (2024), Ali, Gomes et al. (2023), Oravec (2023), Philp et al. (2022), Volberda et al. (2021), Bunker (2020), Kietzmann et al. (2021), Bell et al. (2014)
France	3	2.85	Cloarec et al. (2024), Bunker (2020), Capatina et al. (2020)
Netherlands	3	2.85	Blösser and Weihrauch (2024), Volberda et al. (2021), Overgoor et al. (2019),
Italy	1	1.00	Capatina et al. (2020)
Other European countries	1	1.00	Fu and Fisher (2023)
<i>Asia</i>			
East Asia (China, South Korea and Taiwan)	12	11.42	Bai et al. (2024), Bilal et al. (2024), Cui et al. (2024), Chen and Wei (2024), Kim and Park (2024), Won et al. (2023), Simay et al. (2023), Zhao et al. (2023), Zhu et al. (2023), Ma and Hu (2021), Yang et al. (2021), Bunker (2020),
Southeast Asia (Malaysia, Indonesia)	3	2.85	Yeo et al. (2022), Ameen et al. (2022), Kübler et al. (2020)
India	6	5.71	Haque et al. (2025), Pandey and Vishwakarma (2023), Singh et al. (2023), Bunker (2020), Meena and Kumar (2022), Kushwaha et al. (2021)
<i>Africa</i>			
North Africa (Tunisia)	1	1.00	Saad and Choura (2022)
East Africa (Ethiopia)	1	1.00	Kübler et al. (2020)
<i>Oceania</i>			
Australia	3	2.85	Shumanov et al. (2022), Overgoor et al. (2019), Bunker (2020)
Country not reported	28	26.67	Tian et al. (2024), Chan et al. (2023), Nepomuceno et al. (2023), Oc et al. (2023), Pourranjbar and Shokouhyar (2023), Mustak et al. (2023), Airani & Karande (2022), Binder et al. (2022), Chang et al. (2022), Filieri et al. (2022), Johnson et al. (2022), Overgoor et al. (2022), Pandey et al. (2022), López Vargas et al. (2022), Salminen et al. (2022), Smith et al. (2022), Wu et al. (2022), Allal-Chérif et al. (2021), Dwivedi et al. (2021), Kozinets (2022), Pamuksuz et al. (2021), Rutter et al. (2021), Sinha et al. (2020), Wilson-Nash et al. (2020), Ahani et al. (2019), Steinhoff et al. (2019), Lau and Xu, (2017), and Buettner (2017)

Table 3. Industries investigated in AI-SMM research.

Industry	No. of studies	%	Exemplary studies
<i>Fashion</i>			
Apparel	9	11.49	Gutuleac et al. (2024), Shao (2024), Fu and Fisher (2023), Simay et al. (2023), Zhao et al. (2023), Saad and Choura (2022), Sands et al. (2022), Yeo et al. (2022), Cui et al. (2018)
Beauty	4	4.59	Simay et al. (2023), Ameen et al. (2022), Smith et al. (2022), Rutter et al. (2021)
Advertising	4	4.59	Wang et al. (2023), Wu et al. (2022), Hayes et al. (2021), Kietzmann et al. (2021)
Food and Beverage	5	5.74	Won et al. (2023), Meena and Kumar (2022), Philp et al. (2022), Hartmann et al. (2021), Meire et al. (2017)
Retailing	3	3.44	Allal-Chérif et al. (2024), Bilal et al. (2024), Kim and Park (2024)
<i>Hospitality</i>			
Hotels	4	4.59	Ku et al. (2024), Filieri et al. (2022), Kwok and Yu (2013), Ahani et al. (2019)
Healthcare	1	1.14	Xie et al. (2023)
Tourism	4	3.44	Chen and Wei (2024), Li et al. (2023), Buhalis and Sinarta (2019)
Airline	3	3.44	Chang et al. (2022), Gunarathne et al. (2022), Aljedaani et al. (2022)
<i>Service</i>			
Banking	1	1.14	Shumanov et al. (2022)
Finance	2	2.29	Kozinets and Seraj-Aksit (2024), Wang et al. (2020)
Gaming	1	1.14	Nepomuceno et al. (2023)
Other industries	15	17.24	Bai et al. (2024), Cui et al. (2024), Hu et al. (2023), Pourranjbar and Shokouhyar (2023), Airani and Karande (2022), Binder et al. (2022), Overgoor et al. (2022), Pandey et al. (2022), Salminen et al. (2022), López Vargas et al. (2022), Zhu et al. (2022), Van Esch et al. (2021), Volberda et al. (2021), Kübler et al. (2020), Obschonka et al. (2020)
The industry not explicitly stated	28	33.33	Bell et al. (2014), Blösser and Weihrauch (2024), Brüns and Meißner (2024), Cloarec et al. (2024), Oh and Ki (2024), Tian et al. (2024), Xu et al. (2024), Oravec (2023), Pandey and Vishwakarma (2023), Singh et al. (2023), Johnson et al. (2022), Allal-Chérif et al. (2021), Dwivedi et al. (2021), Van Esch and Stewart Black (2021), Kozinets (2022), Kushwaha et al. (2021), Yang et al. (2021), Ma and Hu (2021), Pamuksuz et al. (2021), Bunker (2020), Capatina et al. (2020), Wilson-Nash et al. (2020), Sinha et al. (2020), Overgoor et al. (2019), Steinhoff et al. (2019), Fang and Hu (2018), Moncrief (2017), Buettner (2017)

While single-nation research provides few, less comparable insights, the majority of studies ($n=28$) concentrate on global variables across many countries. Given the exponential global growth of AI in SMM, it is imperative to comprehend how cultural variations impact consumer behavior. This draws attention to a significant area of unexplored cross-cultural dynamics research.

Table 3 outlines the industries investigated to show how AI in SMM strategies differ across sectors or evolve. In terms of industry contextualization (Table 3), the review reveals that, out of the sixty studies that concentrate on particular industries, the fashion industry is the one that has been studied the most, with 13 studies falling under this category. Research on the fashion sector, including the apparel and beauty market, has become increasingly important. Studies in this sector have grown substantially during the last five years. Four of these concentrated on the beauty market (e.g., Simay et al., 2023; Smith et al., 2022) and ten of them focused on the various apparel markets (Oc et al., 2023; Sands et al., 2022; Shao, 2024; Zhao et al., 2023).

A range of sub-industries is included in the hospitality industry, including hotels (4 studies; Ku et al., 2024; Ahani et al., 2019), tourism (3 studies; Chen & Wei, 2024; Buhalis & Sinarta, 2019). Since the hospitality sector is so important, more studies are needed to determine whether previous findings can be applied to this new context. Similarly, the healthcare industry is covered by only one research study (Xie et al., 2023), which focuses on patient involvement and trust-building. Given the sensitivity of this business, AI solutions are used with caution and a focus on ethical issues and data privacy. The other service sectors, which comprise banking services (Shumanov et al., 2022), finance services (Kozinets & Seraj-Aksit, 2024), and gambling services (Nepomuceno et al., 2023), account for 4.59% of the total (four studies). Furthermore, only three studies (e.g., Allal-Chérif et al., 2024; Kim & Park, 2024) focused on the retailing sector showing stronger emphasis on AI technologies as well as personalized marketing through AI. In comparison, 16 research focus on other industries (such as the manufacturing and residential housing markets), and other studies (Bunker, 2020; Capatina et al., 2020) do not analyze any industry at all.

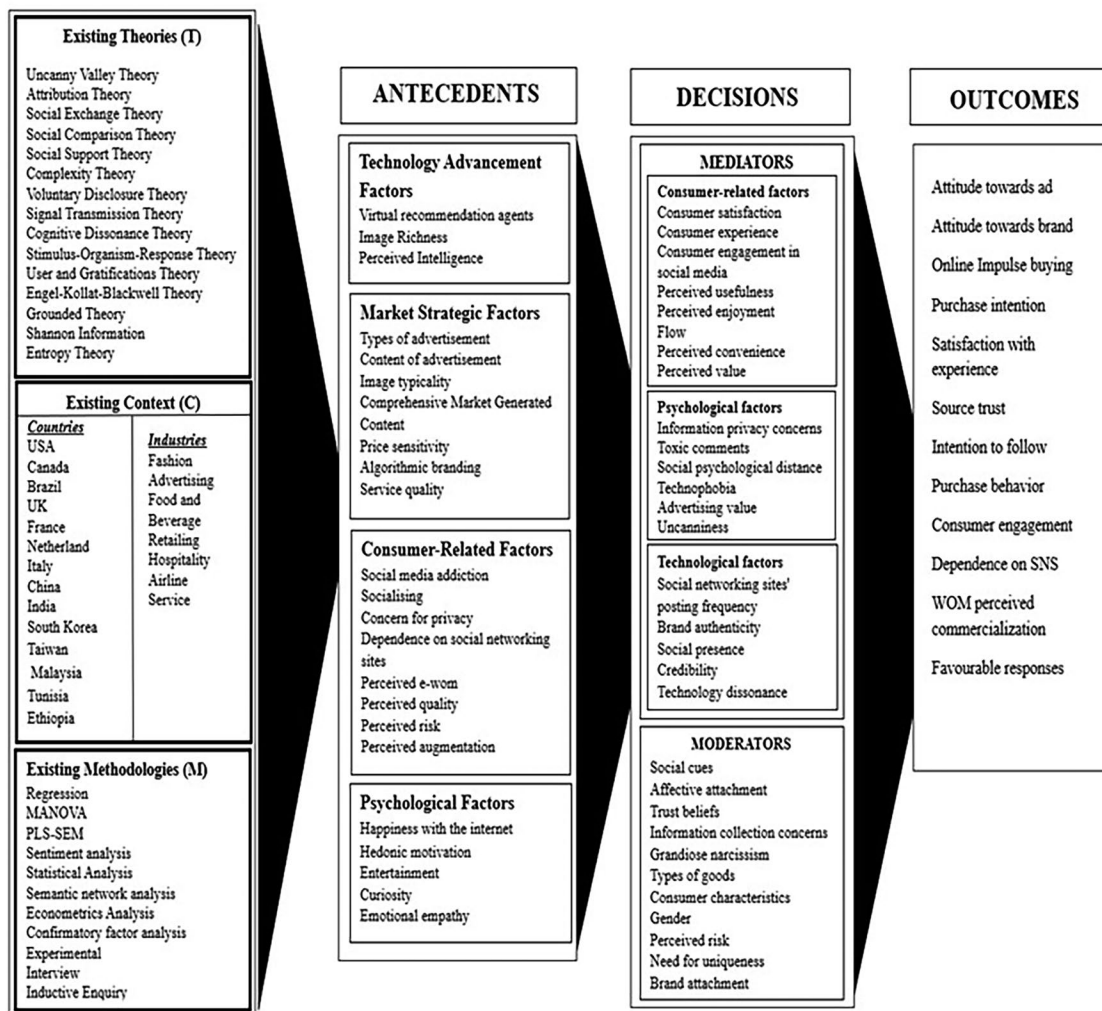


Figure 2. Conceptual framework of AI-SMM.

Characteristics

Through the antecedents-decisions-outcomes (ADO) stages, researchers have looked at a variety of business, consumer, and competitor characteristics in the context of AI in SMM. In order to determine the themes that guided the conceptual model, we used a systematic process. In the first instance, following the collection and screening of the original batch of articles, each article was carefully reviewed, and relevant areas of the study were grouped using an inductive approach as per their commonalities and conceptual linkages. The ADO framework, which offered an organizational structure for grouping research ideas into antecedents, decisions, and outcomes, served as the basis for this procedure. The variables were identified by two researchers by reviewing all the 89 articles and then according to the direction of the studies, similar variables were grouped together according to

ADO framework. After that we consulted the professionals for naming of the themes, and then finalized four main themes for antecedents and three for decisions (mediators).

Based on directions of the reviewed studies, the themes are listed with their variables in Figure 2 as independent, mediating, moderating, or dependent. An analysis of 89 articles reveals four categories with a total of 23 antecedents, three categories of decisions (mediators) with a total of 19 mediator's variable and moderators which are also part of decisions includes 11 variables and for outcomes, there were 12 variables. By seeing decisions as reactions to antecedents and initiatives toward future occurrences, the ADO framework examines these traits (e.g., Maru & Sai Vijay, 2024). The decision in this study is represented by AI in SMM, with antecedents being the elements that influence it and outcomes being

those that are impacted by it. The framework describes the causes, effects, and coping strategies of AI in SMM.

Antecedents: The “antecedents” relate to the factors that have facilitated the advancement and widespread usage of AI technology in SMM. In this paper, factors related to technological advancements, consumer behavior, market strategies, and psychological factors are identified. Consequently, these antecedents were grouped into four different categories. So, factors related to technology advancements include the factors such as virtual recommendation agents (Saad & Choura, 2022), Image richness (Zhao et al., 2023), Perceived intelligence (Shao, 2024). Likewise, the factors related to the market strategy include the choices and the steps taken by the marketers or businesses to achieve their marketing objectives. These factors have influenced the decision to use AI in SMM. They include Types of advertisement, content of the advertisement (Dwivedi et al., 2021), Image typicality (Philp et al., 2022), Comprehensive marketer-generated content (MGC) (Nepomuceno et al., 2023), Algorithmic branding (Kozinets, 2022), and Service quality (Shao, 2024). Whereas the consumer-related factors include all those antecedents which help consumers in decision making and these include Perceived e-WOM, Perceived Price, Perceived Quality, Perceived Emotions, Perceived Risk (Yeo et al., 2022), Perceived augmentation (Ameen et al., 2022), Socializing (Wilson-Nash et al., 2020), privacy concern (Wang et al., 2023), Dependence on social networking sites (Wang et al., 2023). Whereas psychological factors include Happiness with the internet (Cloarec et al., 2024), Hedonic motivation (Shao, 2024), Entertainment, Curiosity, and Emotional empathy (Wilson-Nash et al., 2020).

Decisions: Decisions are processes or behaviors that are influenced by preceding events. Decisions are influenced by various mediating and moderating factors. As Paul & Feliciano-Cestero, (2021) noted, moderators affect the strength or direction of the relationship between two variables, while mediators help explain the underlying process linking them. In the context of AI, social media marketing attributes act as

mediators in two ways. Either SMM attributes are direct outcomes of the AI-related variables, which further induce different consumer behavior, or these SMM attributes act as the precursors to the AI variables. It is important to understand the relationships between antecedents or independent factors and outcomes or dependent variables using mediating variables. Eighteen articles found 20 mediating variables related to AI in SMM. These mediators are also classified into three categories: consumer-related, psychological factors and technological-related. So, the mediators related to consumers include Consumer Satisfaction, Consumer Engagement on social media, Consumer Experience (Bilal et al., 2024), Perceived usefulness, Perceived enjoyment, Flow (Kim & Park, 2024), Perceived convenience (Wang et al., 2023), perceived value (Dwivedi et al., 2021). Likewise, psychological factors include Information privacy concerns (Wang et al., 2023), Toxic comments (Nepomuceno et al., 2023), Social psychological distance (Sands et al., 2022), Technophobia (Sinha et al., 2020), Advertising value (Dwivedi et al., 2021), Uncanniness (Gutuleac et al., 2024). Whereas, technological factors include social networking sites (SNS) posting frequency (Cloarec et al., 2024), Brand authenticity (Brüns & Meißner, 2024), Social presence, Credibility (Kim & Park, 2024), Technology dissonance (Wang et al., 2023).

Moderators may alter the link between antecedents and outcomes, shaping decision-making (Figure 2). AI in SMM studies utilize several moderators, including social cues (Gutuleac et al., 2024), affective attachment (Bilal et al., 2024), trust beliefs, brand attachment, brand awareness, and reputation (Ameen et al., 2022), which are psychological and cognitive variables. Other sociological, technological, and economic variables include narcissism types, consumer characteristics, gender, perceived risk (Saad & Choura, 2022), chatbot support (Ameen et al., 2022), product type, fan number, text length (Zhao et al., 2023), trendiness, and biometrics (Van Esch et al., 2021).

Outcomes: The findings highlight the impact of consumers’ behavioral reactions (Paul & Benito, 2018). Numerous outcomes of AI in SMM at the

Table 4. Methodologies used in the prior studies.

Methods	No. of articles	Exemplary studies
Quantitative Research		
Experiment	16	Blösser and Weihrauch (2024); Brüns and Meißner (2024), Cui et al. (2024), Gutuleac et al. (2024), Oh and Ki (2024); Chan et al. (2023), Pandey and Vishwakarma (2023), Ameen et al. (2022), Gunarathne et al. (2022), Overgoor et al. (2022), Sands et al. (2022), Salminen et al. (2022), López Vargas et al. (2022), Wang et al. (2020), Meire et al. (2017), and Buettner (2017)
Regression analysis	5	Bai et al. (2024), Chen and Wei (2024), Wang et al. (2023), Overgoor et al. (2022), and Philp et al. (2022)
Analysis of variance	4	Blösser and Weihrauch (2024), Cui et al. (2024), Gutuleac et al. (2024), and Kwok and Yu (2013)
Causal analysis	1	Tian et al. (2024)
Multivariate Analysis of Variance (MANOVA)	1	Brüns and Meißner (2024)
Confirmatory factor analysis (CFA)	1	Simay et al. (2023)
Machine learning	6	Fu and Fisher (2023), López Vargas et al. (2022), Zhu et al. (2023), Pamuksuz et al. (2021), Wang et al. (2020), and Cui et al. (2018)
Structural equation model	4	Chen and Wei (2024), Kim and Park (2024), Simay et al. (2023), and Wang et al. (2023)
PLS-SEM	1	Bilal et al. (2024)
Vector Autoregressive Models	1	Kübler et al. (2020)
Econometrics Analysis	2	Zhao et al. (2023) and Kübler et al. (2020)
Bayesian network (BN) analysis	1	Airani and Karande (2022)
Adaptive Neuro-Fuzzy Inference System (ANFIS)	1	Ahani et al. (2019)
Support Vector Regression (SVR)	3	Singh et al. (2023), Li et al. (2023), and Ahani et al. (2019)
Statistical Analysis	5	Fu and Fisher (2023), López Vargas et al. (2022), Kübler et al. (2020), Obschonka et al. (2020), and Fang and Hu (2018)
Semantic network analysis	1	Won et al. (2023)
Aspect-based sentiment analysis (ABSA)	1	Chang et al. (2022)
Graph Convolutional Networks (GCN)	1	Hu et al. (2023)
Convolutional Neural Networks (CNNs)	2	Hartmann et al. (2021) and Rutter et al. (2021)
Process Macro	1	Cloarec et al. (2024)
BERT	3	Hu et al. (2023), Binder et al. (2022), and Pamuksuz et al. (2021)
Linguistic Inquiry and Word Count (LIWC)-Total	1	Hayes et al. (2021)
Qualitative Research		
Exploratory	1	Ma and Hu (2021)
Critical review	2	Mustak et al. (2023) and Pandey et al. (2022)
Content analysis	2	Kushwaha et al. (2021) and Yang et al. (2021)
Case study	2	Allal-Chérif et al. (2024) and Allal-Chérif et al. (2021)
Sentiment analysis	4	Xu et al. (2024), Pourranjbar and Shokouhyar (2023), Kushwaha et al. (2021), and Wu et al. (2022)
Delphi method	1	Pourranjbar and Shokouhyar (2023)
Thematic analysis	2	Haque et al. (2025) and Filieri et al. (2022)
Interview	2	Allal-Chérif et al. (2024), Wilson-Nash et al. (2020)
Meta-analysis	1	Dwivedi et al. (2021)
Narrative analysis	1	Oravec (2023)
Comprehensive analysis	1	Steinhoff et al. (2019)
fuzzy-set Qualitative Comparative Analysis (fsQCA)	1	Shao (2024)
Netnography	1	Kozinets and Seraj-Aksit (2024)
Inductive Enquiry	1	Buhalis and Sinarta (2019)
Total	22	
Mixed-method approach (Qualitative cum Quantitative)	12	Ali, Abdelbaki et al. (2023), Nepomuceno et al. (2023), Aljedaani et al. (2022), Meena and Kumar (2022), Shumanov et al. (2022), Smith et al. (2022), Yeo et al. (2022), Sinha et al. (2020), Van Esch et al. (2021), Capatina et al. (2020), Overgoor et al. (2019), and Oc et al. (2023)
Conceptual approach	3	Kozinets (2022), Volberda et al. (2021), and Kietzmann et al. (2021)
Total	15	
Grand Total	98	

consumer level were found in several studies and these include Purchase intention (Bilal et al., 2024; Dwivedi et al., 2021; Smith et al., 2022), purchase behavior (Ameen et al., 2022; Chen & Wei, 2024; Gutuleac et al., 2024), attitude toward add; attitude toward brand (Dwivedi et al., 2021), Satisfaction with experience (Kim & Park, 2024), Online impulse buying (Saad & Choura, 2022),

Source trust; intention to follow; WOM perceived commercialization (Sands et al., 2022), Consumer engagement (Zhao et al., 2023), Dependence on SNS (Wang et al., 2023), Cloarec et al. (2024), As AI enables more personalized interactions, these outcomes help customers feel a stronger connection to brands, which boosts loyalty and satisfaction.

Methodology

Table 4 demonstrates the different kinds of methodologies adopted in the literature reviewed. Most of the studies (77%), as demonstrated by the findings of the literature reviewed, used quantitative methods. Sentiment analysis, content analysis, case studies, and literature reviews were among the qualitative methodologies (22) used in a few investigations. Contrarily, in a small number of studies (11), a mixed methods approach was used, combining quantitative and qualitative procedures. The set of quantitative studies includes 16 papers that use primary experimental data. Depending on the data used, researchers in social media and AI studies use a variety of analytical techniques (see Table 4). As seen in six studies, machine learning was the most often used approach, reflecting an excess of quantitative datasets.

Machine learning techniques like Support Vector Regression (SVR) (Ahani et al., 2019; Singh et al., 2023), BERT (Binder et al., 2022; Hu et al., 2023), Convolutional Neural Networks (Hartmann et al., 2021; Rutter et al., 2021), Graph Convolutional Networks (Hu et al., 2023), Bayesian network analysis (Airani & Karande, 2022) and the Vector Autoregressive Models (Kübler et al., 2020) are frequently used in studies centered around AI and social media. These methods are used to elucidate the motives behind why certain individuals are increasingly utilizing AI in SMM. Studies centered on survey data employ the second most commonly used set of methodologies. Among the mean comparison techniques utilized in this category are repeated measure ANOVA, SEM and multiple analyses of variance (Chen & Wei, 2024; Gutuleac et al., 2024; Kim & Park, 2024; Oc et al., 2023). These conventional fluctuation methods are used more frequently than those that are more advanced. These include, for example, (1) Sentiment analysis, (2) Econometrics analysis, (3) Causal analysis, and (5) Statistical analysis. Remarkably, no single analytical procedure is intrinsically better than any other. Several research uses multiple approaches to manage various kinds of datasets from different sources (Meena & Kumar, 2022; Nepomuceno et al., 2023; Smith et al., 2022) to minimize common method variance and improve

the generalizability of their findings (Podsakoff et al., 2003).

Future research agenda

Numerous studies over the last decade have contributed to better comprehending AI in SMM (Table 5). While attempts have been made to incorporate the theoretical and empirical aspects, a full evaluation has been missing. This review presents an overview of AI in SMM research, looking at the theories employed, the contexts investigated, the characteristics researched, and the methodologies used to analyze the relationships involved.

Theory

Using several theoretical frameworks can help future research on AI usage in SMM obtain a deeper understanding. Even though several researchers have looked at AI in SMM from different perspectives, this technique is still not widely used. Although 14 theories address this problem, few provide thorough explanations, according to the systematic review, which emphasizes the need for greater in-depth theoretical investigation. Given the complexity of AI in SMM, the majority of research only uses one theory, which may not be enough. Building on the theoretical underpinnings will improve comprehension and give this developing field a stronger framework.

By describing how people perceive and respond to their surroundings, psychological theories such as social exchange and social comparison theory (Blau, 1964) contribute to predicting thoughts, feelings, and behavior. Understanding how AI-driven strategies like tailored rewards affect customer engagement is made easier by the self-determination theory (1985), which evaluates both internal and external incentives. By examining how social media users categorize themselves, social identity theory illuminates the impact that targeted marketing, brand communities, and loyalty contribute to group identification. Behavioral theories that concentrate on how behavior is learnt and shaped in AI-SMM include SOR (Mehrabian & Russell, 1974), Uncanny Valley (Mori, 1970), and EKB (Engel et al., 2001).

Table 5. Future research agenda.

Approach	Research gap identified	Future research directions	Future research questions
Theory Development (T)	Relying excessively on traditional psychological and behavioral theories (such as SOR, Social Comparison, and Uncanny Valley).	Create hybrid frameworks that combine psychological, technical and social viewpoints to explain AI in SMM interactions.	In what ways can psychological, technical, and social perspectives be integrated to form a hybrid framework that explains consumer interactions with AI in social media marketing?
	Lack of integrative theoretical frameworks that incorporate cognitive, emotional, and social elements.	Future research should create integrative frameworks that integrate social, emotional, and cognitive components to offer a more comprehensive knowledge of consumer behavior and the mechanisms	How can the integration of algorithm aversion and human-AI teaming theories advance a hybrid framework for explaining consumer trust and engagement with AI in social media marketing?
	Minimal theoretical exploration of AI's ethical and societal implications in SMM.	Investigate cross-theoretical models that combine Uncanny Valley and Attribution theories to study emotional-judgmental reactions to AI influencers or synthetic content.	How can a cross-theoretical model integrating Uncanny Valley Theory and Attribution Theory explain consumers' emotional and judgmental responses to AI influencers and synthetic content in social media marketing?
Context (C)	The majority of studies employ old theories and fail to take global culture and ethics into account when discussing AI and social media marketing.	Develop new models that combine these concepts to assist marketers in anticipating and comprehending customer responses in various countries.	How can we better understand how people around the world react to AI in marketing by integrating concepts from psychology, culture, technology, and ethics?
	The dominant focus is on developed economies, particularly North America and portions of Asia.	Conduct cross-national research comparing individualistic and collectivistic cultures to investigate the cultural influences on AI in SMM efficacy.	What effects do cultural settings, individualist versus collectivistic cultures in particular, have on customer behavior and the efficacy of AI-powered social media marketing?
	Emerging and developing economies are under-represented (for example, Africa, South America, and Eastern Europe).	Investigate AI-SMM in underserved sectors such as public healthcare, rural banking, education technology, and small-scale retail in developing countries.	What sector- and context-specific factors affect the adoption and efficacy of AI-driven social media marketing in underserved sectors in developing nations, such as small-scale retail, public healthcare, rural banking, and education technology?
Characteristics (C)	An overemphasis on the fashion, tourism, and hospitality sectors, while important businesses such as healthcare, education, public services, and banking are underexplored. Numerous studies merely look at people's opinions rather than their actual behavior or social media usage around the globe.	Investigate AI applications in politically sensitive or regulated industries to better understand the impact of policy, data protection, and ethical governance.	How do ethical governance, data protection, and policy affect the application and efficacy of AI-driven social media marketing in highly regulated or politically sensitive sectors?
	Excessive dependence on perceived factors (risk, price, etc.) without considering real-world behaviors.	Examine actual behaviors and the ways that new social media trends affect trust and loyalty in a variety of international industries.	What effects do privacy concerns and actual social media use have on consumers' trust and loyalty to AI-using firms in various nations's?
	Limited integration of SNS engagement indicators like posting frequency and social presence.	Move beyond perceptions to actual usage and behavior patterns by assessing the role of data privacy and SNS dependence.	How do characteristics such as data privacy concerns and social media platform dependence influence actual usage patterns and consumer behavior in AI-driven social media marketing?
Methodology (M)	Focuses largely on traditional outcomes such as purchase intent and satisfaction	Examine changing SNS trends and how they mediate AI content reception.	How do evolving social networking site (SNS) trends and features, like image richness, service quality, algorithmic branding and content formats, affect how consumers respond to and interact with AI-powered social media marketing content?
	The majority of research is conducted in wealthy nations and a small number of industries, leaving out other nations and sectors around the globe.	Analyze the formation of trust, everlasting loyalty, and the effects of over-personalization and privacy invasion.	What effects do traits of AI-driven personalization—like perceived intelligence, perceived augmentation, and perceived privacy invasion have on building long-term customer loyalty and trust in social media marketing?
	Major emphasis on quantitative approaches (mostly surveys and experiments), qualitative and mixed-method research is underrepresented. Limited use of econometric, ethnographic, and critical review methodologies.	To better understand local variances, do more studies in other sectors and in different regions.	What effects do various sectors, regulations, and cultures have on how social media marketing and artificial intelligence operate in various nations?
		Use mixed-method approaches to improve triangulation and verify AI in SMM components across a variety of datasets.	In what ways may mixed-method approaches be used to validate the components of AI-driven social media marketing across different data sets and triangulate findings?
		Increase the application of advanced machine learning techniques (e.g., ANFIS, deep learning, fuzzy-set QCA) to get more insights.	How can we better understand AI-driven social media marketing processes and results by employing advanced machine learning techniques like fuzzy-set QCA, deep learning, and ANFIS?
		Conduct econometric analysis to investigate the causal effects of AI.	How can econometric analysis be employed to examine the causal effects of AI-driven interventions on social media marketing outcomes?
	To examine global disparities, few studies use interviews, large data, or mixed approaches; instead, they rely on basic surveys.	To better capture actual marketing effects globally, combine data analysis, interviews, and surveys.	Can employing interviews, large data, and mixed methods helps us in better understanding of AI marketing outcomes in various nations by ?

Through content and targeted advertisements, the Theory of Planned Behavior and Elaboration Likelihood Model provide additional insight into how AI affects decision-making in SMM. The internal states of customers are investigated using cognitive theories such as cognitive dissonance theory (Festinger, 1957) and attribution theory (Folkes, 1988).

Developmental theories like Grounded Theory help foster organizational commitment, promoting a cohesive and motivated workforce. Cultural theories, such as the Cultural Dimensions Theory, examine how cultural differences impact AI in SMM across geographic and demographic segments. Network theories can aid future studies in understanding information flow and influence within social networks, particularly regarding influencer partnerships and viral marketing. Additionally, theories focused on user interaction, engagement, and experience, such as human-computer interaction, communication, and user experience theories are essential for understanding AI's impact on consumer involvement on social media platforms.

Context

Understanding the country and industry contexts is essential to comprehending AI's impact on SMM. These areas have been the main focus of earlier research, including Chen and Wei (2024) and Pandey et al. (2022). Both industry-specific and country-specific factors have a big impact on AI-SMM. The majority of the current study focuses on industries such as hotels (Buhalis & Sinarta, 2019; Filieri et al., 2022), food and beverage (Hartmann et al., 2021; Meena & Kumar, 2022), and fashion (Saad & Choura, 2022; Yeo et al., 2022). Research in other industries is scarce, though. One study in healthcare (Xie et al., 2023), one on retail banking advertisements (Shumanov et al., 2022), and three studies in retail (Allal-Chérif et al., 2024; Bilal et al., 2024; Kim & Park, 2024) and one in gaming (Nepomuceno et al., 2023) discuss AI in SMM. Additionally, social media personas frequently don't correctly represent their true selves, which results in inaccurate profiling (Shumanov et al., 2022).

AI-identified personality traits tend to be more persuasive when used in marketing promises.

Numerous studies emphasize how AI can be used to predict consumer behavior. According to Capatina et al. (2020), the efficacy of AI in SMM depends on its capacity to fulfill its promised features. Nevertheless, some studies ignore contextual elements, producing inconsistent findings. According to Johnson et al. (2022), for instance, integrating human knowledge with AI can solve complicated problems. Future research should look into strategies to lessen AI bias to assure moral and responsible design.

Future studies ought to examine SMM and AI in cross-cultural or cross-national contexts. Numerous studies have been carried out worldwide, but the majority concentrate on developed economies, especially the USA. China (Simay et al., 2023; Zhao et al., 2023), India (Pandey & Vishwakarma, 2023), and Australia (Shumanov et al., 2022) are also conducting significant studies. But regions like Russia, Germany, and New Zealand are still not well studied. The majority of the literature ignores variations in behavior of social media users, market segmentation, and digital marketing across various cultures and economies in favor of concentrating on particular nations (Bilal et al., 2024; Yeo et al., 2022). It may be useful to compare collectivistic and individualistic societies. This discrepancy restricts the findings' generalizability and emphasizes the necessity of cross-national research to pinpoint the consistent and variable elements affecting AI in SMM (Kim & Park, 2024).

Characteristics

According to the study, AI in SMM incorporates market tactics, consumer-related and psychological aspects, and technological developments. The effects of AI-assisted personalization on user pleasure, loyalty, and perceived content relevance remain to be investigated, even if several variables have been looked at as explanatory, controlling, mediating, and standard. It is crucial to incorporate customization and personalization factors. According to Kim and Park (2024), customer satisfaction is a positive reaction to past expectations or experiences; however, further study on consumer satisfaction and trust is required to evaluate the efficacy of AI in SMM. Future research should

also look at how privacy and data security issues affect customer behavior and brand trust in AI-driven marketing. Van Esch et al. (2021) use social media usage as a mediating variable and trendiness and biometrics as moderating variables, but overlook how excessive social media use and privacy concerns affect consumer behavior.

Again, there are some variables like Social presence (Kim & Park, 2024), Uncanniness (Gutuleac et al., 2024), Advertising value (Dwivedi et al., 2021) and Toxic comments (Nepomuceno et al., 2023), used as mediators in past studies. However, they can also be used as a moderator in future studies. Furthermore, mediators like Perceived usefulness, Perceived enjoyment (Kim & Park, 2024), Perceived convenience (Wang et al., 2023), and Perceived value (Dwivedi et al., 2021) are the perceived variables that are taken in past studies. However, future researchers can study the actual effect to increase an in-depth understanding. Further, the themes found in this research are dynamic rather than static. As new research issues arise and as researchers use different theoretical bases, themes should change over time. In this regard, the use of more recent frameworks might offer new interpretations, while expanding concerns about various new future research problems like generative AI, digital trust, and sustainability may expand or change current themes. Moreover, in line with the evolving nature of the identified themes, future research in this domain would find strong alignment with journals in the area of consumer behavior, technology behavior, and marketing.

Methodology

The methodological procedures used in the reviewed studies reveal important issues that future research should address. As AI and SMM research continues to evolve, it is critical to validate findings using a variety of methodologies, sample groups, and longitudinal studies to ensure reliability and validity. The majority of the studied literature ($n=60$) focused on quantitative methods, mainly experiments and multivariate analysis, while just 22 studies used qualitative approaches, revealing a methodological gap. Only two studies

used econometric approaches (Ma & Hu, 2021). According to Brüns and Meißner (2024), future studies could employ econometrics to analyze the impact of AI on user engagement. Furthermore, machine learning-based tools should be prioritized over surveys, and methodologies such as ethnography or content analysis of SMM may provide more in-depth insights into AI's effects.

Theoretical approaches should be expanded to develop the construct, as only three reviewed publications have used this strategy, highlighting the need for greater accessibility. Secondary data collection provides future study opportunities by overcoming the limits of single-nation data and increasing sample numbers, which improves generalizability. Furthermore, a framework-based critical review could be undertaken using paradigms such as Theory-Context-Methods (TCM), Antecedents-Dimensions-Outcomes (ADO), and TCCM (Paul & Criado, 2020) to identify specialized areas within the discipline.

Future studies should conduct meta-analyses to better quantify AI's impact on SMM. While most research relies on traditional methods like regression and SEM, advanced approaches such as the Adaptive Neuro-Fuzzy Inference System (ANFIS), text mining, machine learning, and fuzzy-set qualitative comparative analysis.

The need for more creative and globally aware research in AI-driven social media marketing is emphasized by the future study directions indicated in Table 5. Beyond traditional concepts, researchers should integrate knowledge from psychology, technology, and culture to gain a deeper understanding of how consumers engage with AI in SMM. Analyzing actual consumer behavior, rather than only opinions will provide more precise understanding of how social media and privacy practices impact loyalty and trust. In under-represented fields like healthcare and education, researchers are advised to look beyond well-studied industries and investigate how local culture, laws, and industry types influence AI in marketing. Finally, combining big data analysis, surveys, and interviews will help provide a complete picture of marketing's global influence and guarantee that research findings are reliable and helpful to global marketing executives.

Implications and conclusion

To begin, this systematic review synthesized AI in SMM literature, and the evolution of the AI concept, along with the current state of AI-powered SMM research, has been discussed. The idea of AI in SMM as a potential issue has received limited attention from business and marketing researchers. AI applications used by social media channels include evaluating content and images, identifying spam, promoting, offering social insights, and collecting content.

The study contributes to current theory and has implications for society and the real-world environment. The theoretical underpinnings and research approaches employed in this field are explained by the TCCM-ADO framework. The advent of AI and other current growth have changed SMM, and by investigating future research avenues, the critical review will broaden our understanding regarding the implications of AI in SMM. At the same time, this highlights the requirements for more theoretical models that account for human-AI interaction complexity. The findings from the current SLR demonstrate that AI is considerably changing global marketing practice through social media marketing, with considerable consequences for both academic theory and management implementations. With the help of AI-driven strategies, marketers can forecast consumer behavior, hyper-personalized content, and maximize campaign effectiveness in a variety of foreign countries, resulting in increased global brand loyalty and deeper customer engagement. Given the competitive and fragmented nature of contemporary digital ecosystems, advanced AI tools enable brands to manage swift changes in global market conditions, analyze complex cultural and behavioral data, and customize messaging to local preferences.

Big international companies like Nike, BMW, and Coca-Cola are prime examples of how AI can be used for dynamic personalization, influencers engagement, and real-time social media campaign management. These strategies have a direct impact on customer happiness and worldwide reach. The study does, however, also argue that research gaps in the areas of sectoral inclusion, cross-cultural applicability, and the ethical

use of AI, underscoring the necessity for integrative theoretical frameworks that take into account the reality of international marketing. By filling up these gaps, marketers will be able to properly manage consumer trust, privacy, and brand equity in global contexts by achieving untapped levels of efficiency and return on investment.

Concerning enhanced marketing strategies with an emphasis on personalization, consumer interaction, and targeted advertising, marketers may create more successful AI-driven marketing strategies by utilizing insights from the SLR. Hence, businesses may make more informed decisions and more strategic plans if they have a good knowledge of the benefits and challenges of implementing AI technology within their marketing campaigns. Further, the review can encourage firms to try out new AI methods and advances in technology by stimulating creativity and innovation in marketing strategies. Most importantly, the ADO aids in identifying various antecedents and outcomes of AI in SMM. It may reveal underlying factors related to technological advancements, marketing strategies, and consumer-related and psychological influences, which can create consumer decisions regarding AI in SMM and how this decision further leads to various consumer-related outcomes.

For businesses using AI in social media marketing, the present review has significant ramifications. Businesses may now provide personalized and interactive content, increasing customer engagement and loyalty, thanks to AI-driven technologies like recommendation algorithms and natural language processing. By examining regional customer state of mind, hedonic motivations, and cultural preferences, AI may help global businesses become more cross-culturally adaptive. This enables marketers to customize messaging while preserving brand coherence. By recognizing micro-segments, making the most of scarce resources, and encouraging genuine community-based connections, local companies may use AI to compete more successfully. Additionally, real-time data and sentiment tracking driven by AI provide businesses a way to proactively address customer problems, fostering long-term connections and trust.

The SLR improves knowledge of AI in SMM by integrating and organizing previous research under the lenses of theory, context, characteristics, and methodologies. It identifies significant gaps, such as underserved industries and regions. Present study analyses 89 research to provide a comprehensive framework for demonstrating how AI has transformed social media marketing methods and impacting customer behavior. It is an invaluable resource for researchers looking to strengthen theoretical models, as well as practitioners seeking to adopt more successful, ethical, and consumer-focused AI-driven marketing methods. This presents both potential and problems for practitioners: whereas AI can improve customer engagement, happiness, and personalization, businesses must also deal with privacy issues, perceived risk, and ethical problems. Adoption by the industry should therefore occur with a balance between innovation and transparency, supported by continual monitoring of customer responses and market outcomes. Though there is a lack of extensive empirical evidence across industries, the emerging literature indicates that AI applications like recommendation systems, influencer marketing analytics, and perceived intelligence are starting to change how industries interact with their customers. This presents both opportunities and challenges for practitioners, as while AI can improve personalization, enjoyment, and customer engagement, industries must also deal with ethical issues, perceived risk, and privacy concerns. As a result, industry adoption must continue with a balance of innovation and responsibility, supported by constant surveillance of consumer responses and market outcomes.

Moreover, Despite AI's transformational role in SMM, the literature lacks in-depth consideration of governance, bias, data privacy, and disinformation, all of which are key problems in today's AI world. Policymakers must create legal frameworks to assure ethical AI adoption, increase algorithmic transparency, and protect user autonomy. There is also an urgent need for norms governing influencer marketing, deepfakes, and algorithmic targeting, especially on platforms used by minors or vulnerable groups.

To summarize, this research presented a thorough analysis of a critical review of AI in SMM.

To do this, a detailed investigation of the literature on AI in SMM was conducted. Subsequently, a TCCM paradigm was created to outline the prevailing theories, contextual elements, characteristics and methods in the field. Further, ADO is used to analyze the antecedents, decisions and outcomes of AI in SMM. Following the ultimate objective of this study, enlightening avenues for further investigation are suggested. These approaches provide prospective areas of exploration for future studies in the field of AI and SMM, and the SLR on AI in SMM may greatly help knowledge advancement, policy formation, and future research direction in this quickly developing sector by addressing these results.

While AI is improving SMM's efficiency and personalization on a global basis, there are numerous challenges to overcome in order to understand cross-cultural efficacy, actual user behavior, and ethical conduct. These results ensure a better connection between what the field has learnt and what should be explored next by directly supporting and motivating the research gaps and future directions listed in [Table 5](#).

Limitations

While this review makes substantial contributions to AI in SMM research, there are certain constraints to consider. First, we only considered peer-reviewed academic papers in English that were indexed by Scopus, omitting publications in other languages, which could create bias. Additionally, only journals ranked Q1 on SCImago and A/A* on the ABDC were considered, by the study's quality control standards. Future studies should address these limitations and carry out a more thorough examination to confirm the results. Second, in addition to the TCCM-ADO framework, future studies could consider additional frameworks such as the 5WH or 7P frameworks (Paul et al., 2021), which provide a more objective approach without the necessity for statistical confirmation (Wolf, 2023).

Author contributions

CRedit: **Satinder Kumar**: Conceptualization, Formal analysis, Investigation, Methodology, Supervision, Validation,

Writing – review & editing; **Dipti Malhotra**: Conceptualization, Investigation, Methodology, Validation, Visualization, Writing – original draft; **Garima Kathuria**: Conceptualization, Methodology, Visualization, Writing – original draft.

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